Timing

File: timing.kicad_sch

Sprite Table

File: sprite_table.kicad_sch

Sprite ROM and RAM

File: sprite_rom_and_ram.kicad_sch

Sprite Frame Buffers

File: sprite_frame_buffers.kicad_sch

Timing diagrams

File: timing_diagrams.kicad_sch

Sprite timing diagrams

File: sprite_timing_diagrams.kicad_sch

Spare parts on video PWB(B) 302191A

B_G2C 74LS04

B_G2B 74LS04

B_G2E 74LS04

B_G2D 74LS04

B_H11B 74LS74

B_A11A 74LS32

Spare parts are set as NC.

Spare parts on main PWB(A) 302190A

A_L2A 74LS00

A_L2D 74LS00

A_H8A 74LS07

A_H8D 74LS07

A_K1C 74LS32

A_L1C 74LS07

A_L1B 74LS07

A_L8F 74HC367

A_N3B 74LS139

GND

KiCad does not support reference designators starting with a number. Therefore the references used in the original schematics have been mirrored. 6R -> R6.
The original schematics had separate pages for the main and video PCB. They are combined here. This makes some references duplicate. To distinguish between boards A_ is added before part positions on PWB(A) and for PWB(B) B_ and so on. If there are still conflicts a "P" (Position) is added after board letter AP_, BP. Sometimes there are duplicate reference designators then "CAP" maybe added after board letter ACAP_ (CAP is for capacitor).

Tile Layers

File: tile_layers.kicad_sch

Color mixer

File: color_mixer.kicad_sch

IO

File: io.kicad_sch

Sound

File: sound.kicad_sch

Sound mixer

File: sound_mixer.kicad_sch

Main CPU

File: mcpu.kicad_sch

Shared Resources

File: shared_resources.kicad_sch

Slave CPU

File: scpu.kicad_sch

Fixed Layer

File: fixed_layer.kicad_sch

Only used on GX903 – Cue Brick

Battery Powered RAM

File: bpram.kicad_sch

There are several games and variations on the twin 16 platform:
- Main Board PWB(A)
- Video Board PWB(B)
- Additional ROM Board, if not ROM's mounted directly on video board.
- Battery powered RAM board

Id	PWB(A)	PWB(B)	PWB(C)
Dark Adventure	GX687	302190A	302191A –
Gradius 2	GX785	302190A	302191A –
Vulcan Venture	GX785	302190A	302191A –
Gradius 2	GX785	302190B	302191B 402145A
Hard Puncher	GX870	302190B	302191B 402145A
M.I.A	GX808	302190B	302191B 402145A
Cue Brick	GX903	302190B	302191B –

Cue Brick has an additional Battery Powered RAM Board:
PWB451803

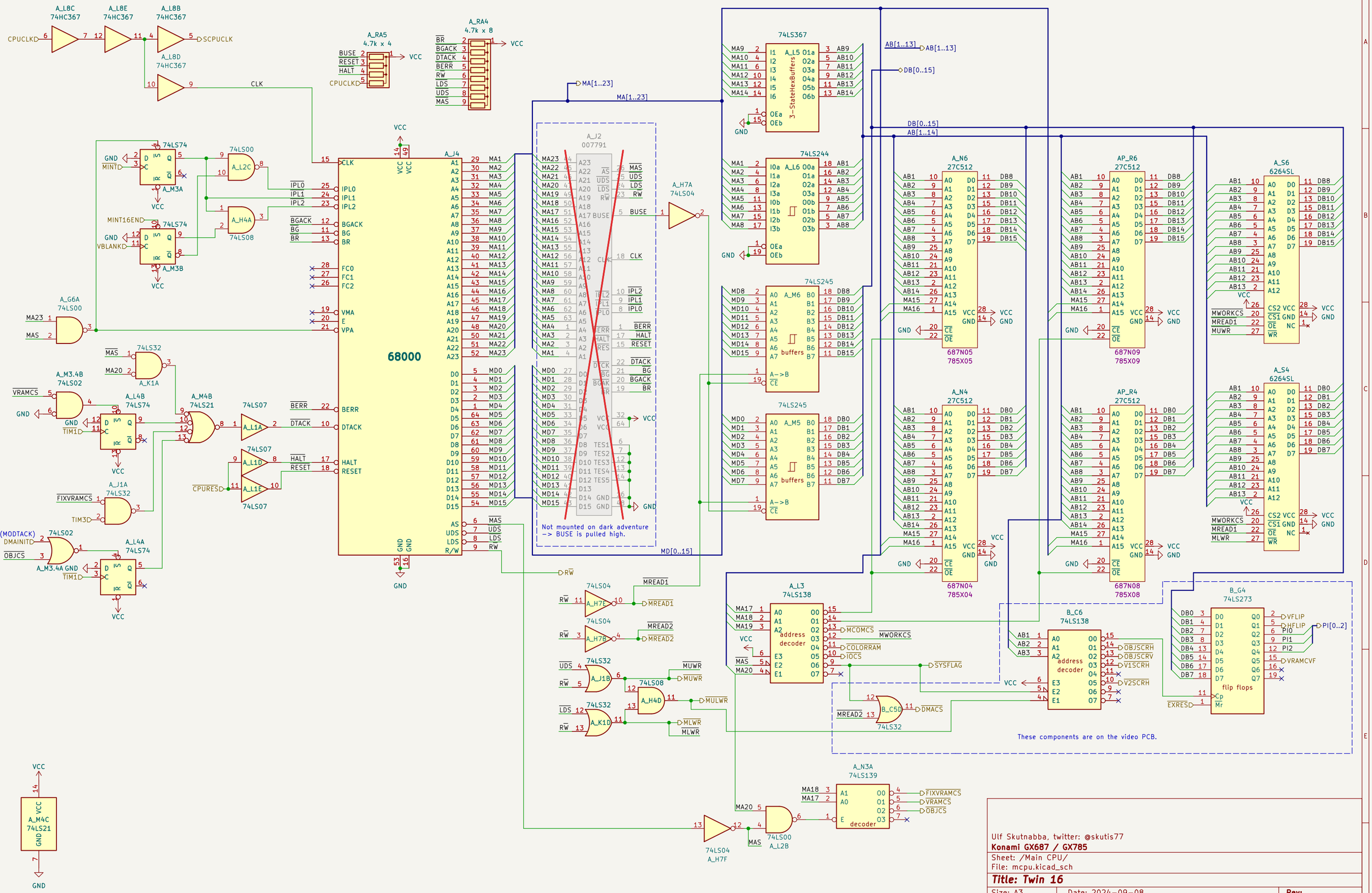
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /
File: darkadv.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/17



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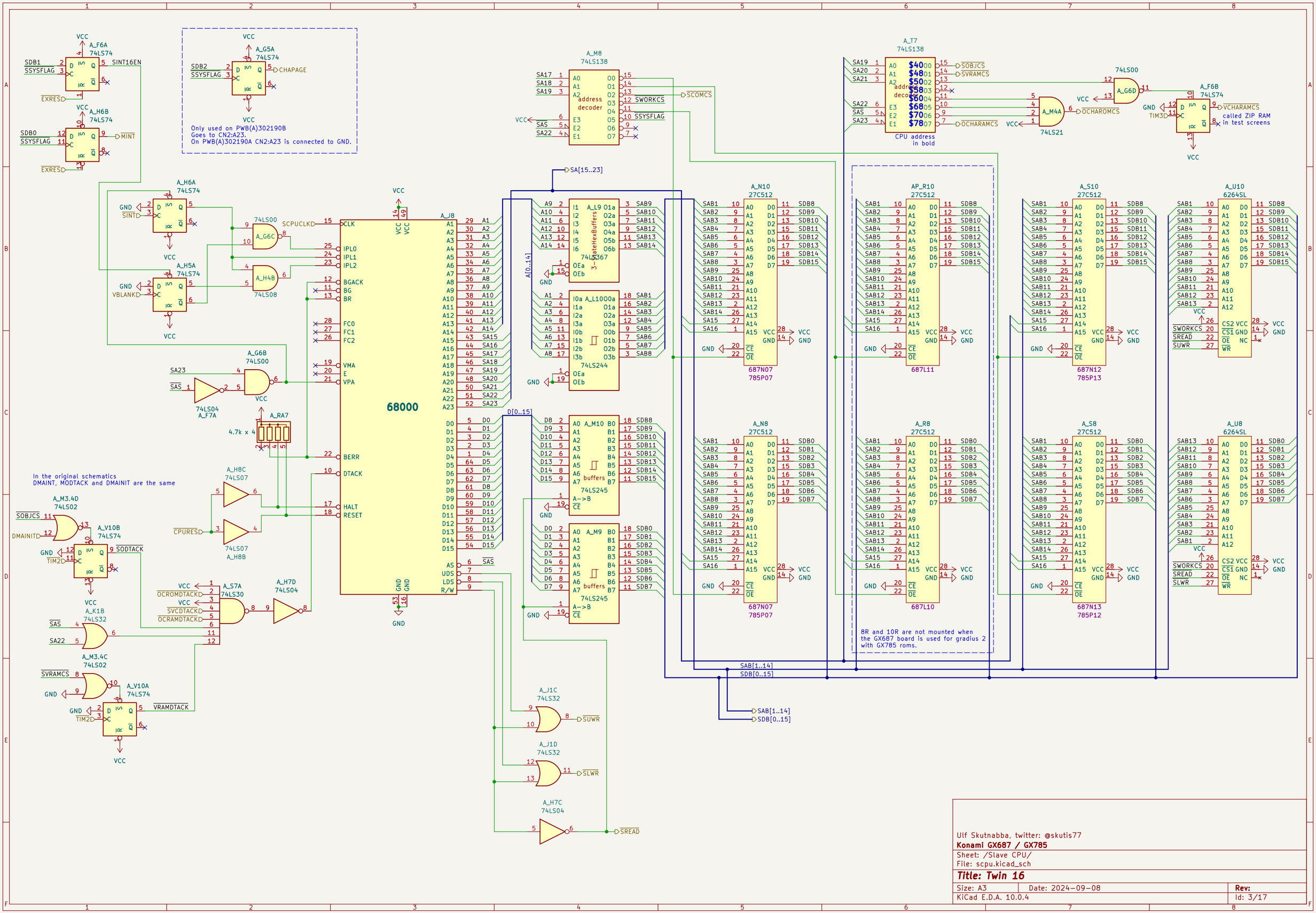
Konami GX687 / GX785

Sheet: /Main CPU/

File: mcpu.kicad_sch

Title: Twin 16

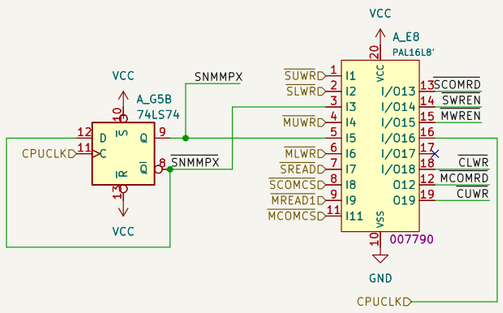
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/17

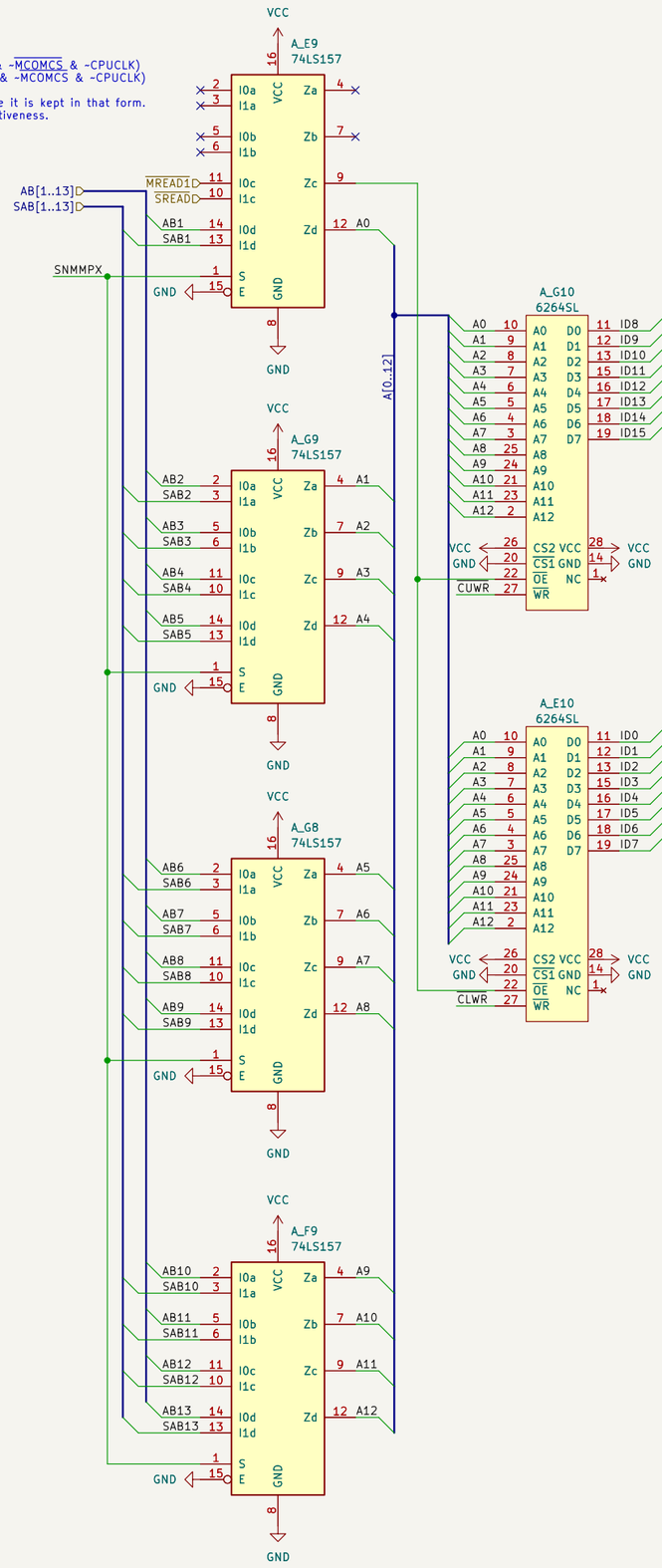

A_F6A 74LS74
SDB1
SSYSFLAG
SINT16EN
EXRES
A_H6B 74LS74
SDB0
SSYSFLAG
MINT
EXRES
A_G5A 74LS74
SDB2
SSYSFLAG
CHAPAGE
Only used on PWB(A)302190B
Goes to CN2:A23.
On PWB(A)302190A CN2:A23 is connected to GND.
A_M8 74LS138
SA17
SA18
SA19
SAS
SA22
address decoder
SCOMCS
SWORKCS
SSYSFLAG
A_T7 74LS138
SA19
SA20
SA21
SA22
SAS
SA23
SOBJCS
SVRAMCS
OCHARAMCS
OCHAROMCS
CPU address in bold
74LS00
A_G6D
A_M4A
74LS21
A_F6B 74LS74
VCHARAMCS
called ZIP RAM in test screens
TIM3
SA[15..23]
A_H6A 74LS74
SINT
A_H5A 74LS74
VBLANK
74LS00
A_G6C
A_H4B
74LS08
SCPUCLKD
68000
CLK
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
AS
UDS
LDS
R/W
A[0..14]
D[0..15]
A_L9 3-StateHexBuffers 74LS367
A_L10 74LS244
A_M10 buffers 74LS245
A_M9 buffers 74LS245
A->B
A_N10 27C512
687N07
785P07
AP_R10 27C512
687L11
A_S10 27C512
687N12
785P13
A_U10 6264SL
A_N8 27C512
A_R8 27C512
687L10
A_S8 27C512
687N13
785P12
A_U8 6264SL
8R and 10R are not mounted when the GX687 board is used for gradius 2 with GX785 roms.
SAB[1..14]
SDB[0..15]
A_G6B 74LS00
SA23
SAS
74LS04 A_F7A
A_RA7
4.7k x 4
In the original schematics
DMAINT, MODTACK and DMAINIT are the same
A_M3.4D 74LS02
SOBJCS
DMAINIT
A_V10B 74LS74
SODTACK
TIM2
A_S7A 74LS30
OCROMDTACK
SVCDTACK
OCRAMDTACK
A_H7D 74LS04
A_K1B 74LS32
SA22
A_M3.4C 74LS02
SVRAMCS
A_V10A 74LS74
VRAMDTACK
A_H8C 74LS07
CPURESD
A_H8B
A_J1C 74LS32
SUWR
A_J1D 74LS32
SLWR
A_H7C 74LS04
SREAD
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Slave CPU/
File: scpu.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
KiCad E.D.A. 10.0.4
Rev:
Id: 3/17

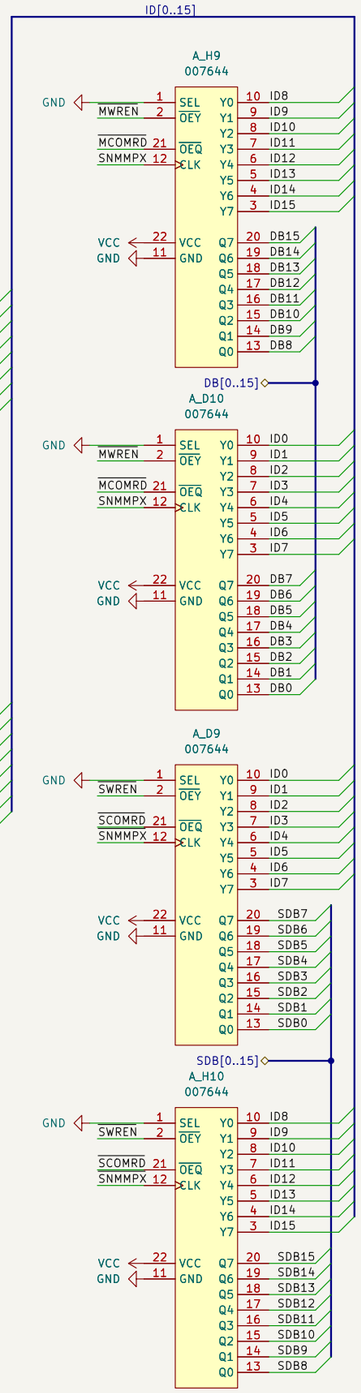
Konami 007790

$\overline{MCOMRD}$ = ~($\overline{MREAD1}$ & ~$\overline{MCOMCS}$) = $\overline{MREAD1}$ | $\overline{MCOMCS}$
$\overline{SCOMRD}$ = ~($\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{SWREN}$ = ~(~SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{MWREN}$ = ~(~$\overline{SNMMPX}$ & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$)
$\overline{CLWR}$ = ~(~$\overline{SLWR}$ & ~$\overline{SNMMPX}$ & SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$ & ~CPUCLK | ~$\overline{SNMMPX}$ & ~$\overline{MLWR}$ & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$ & ~CPUCLK)
CUWR = ~(~SUWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~MUWR & ~SNMMPX & MREAD1 & ~MCOMCS & ~CPUCLK)

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form.
The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.







This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs
mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs.
If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

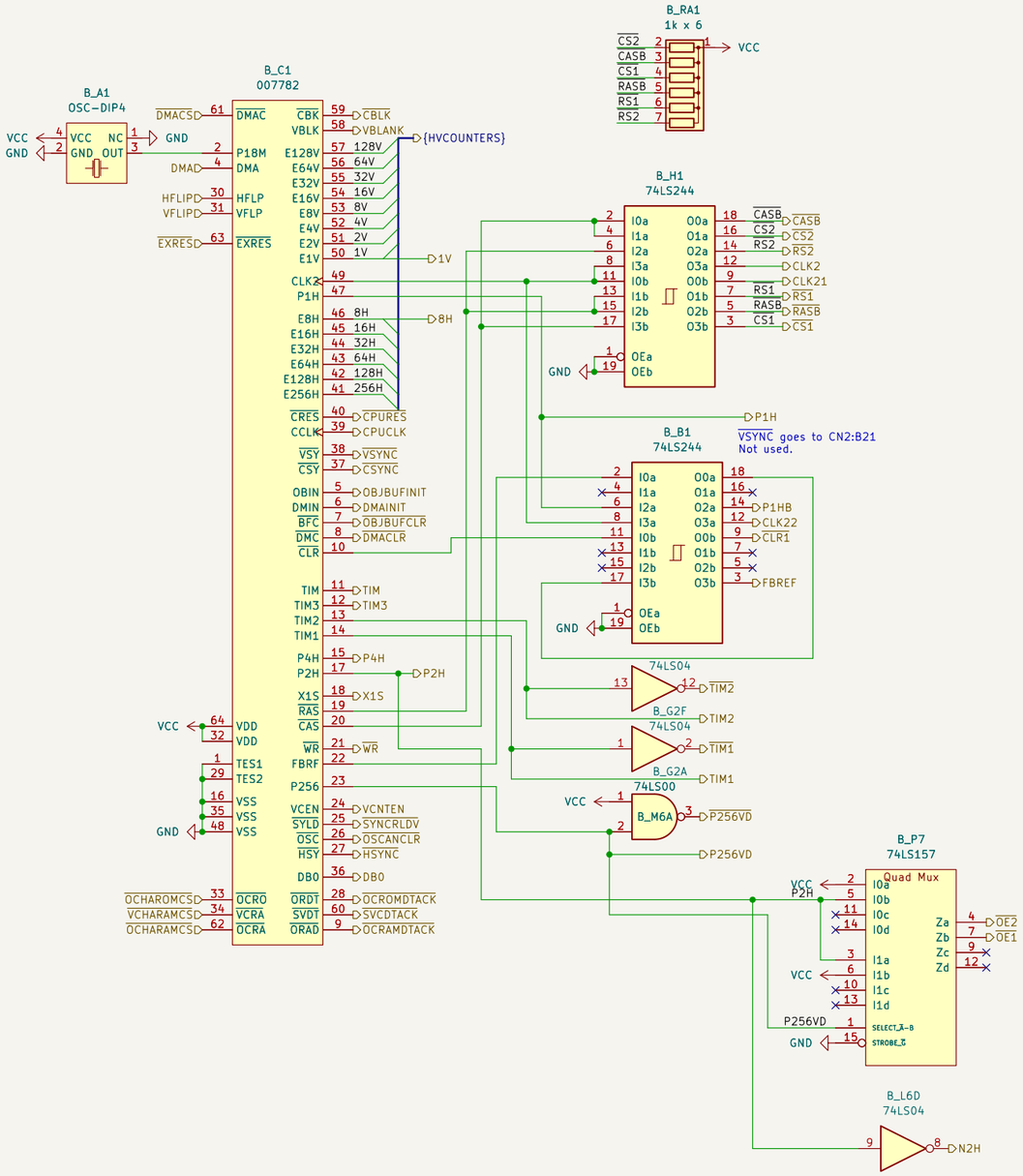
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

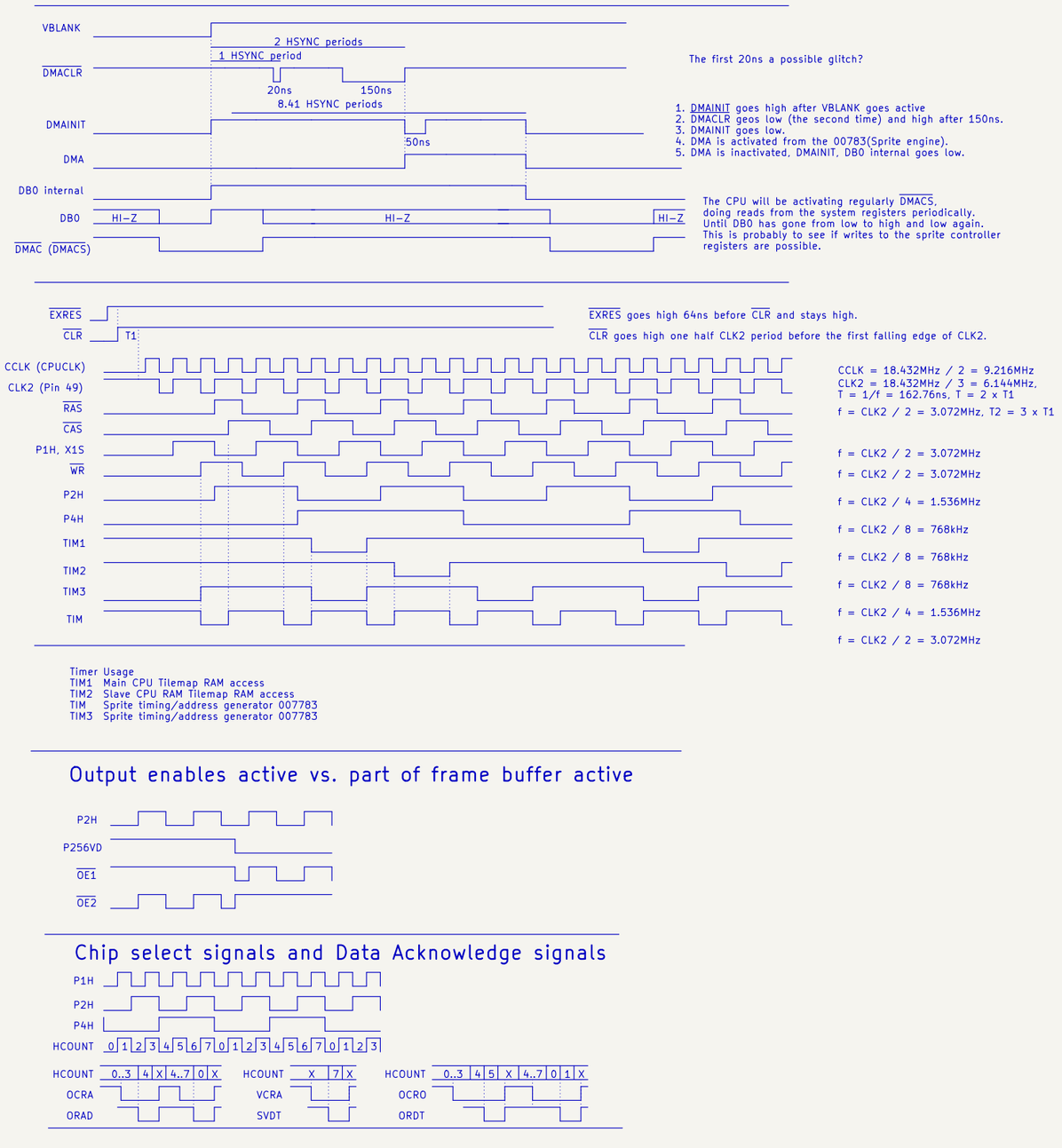
Sheet: /Shared Resources/
File: shared_resources.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/17



DMA Synchronization



Timer Usage
TIM1 Main CPU Tilemap RAM access
TIM2 Slave CPU RAM Tilemap RAM access
TIM Sprite timing/address generator 007783
TIM3 Sprite timing/address generator 007783

Output enables active vs. part of frame buffer active

Chip select signals and Data Acknowledge signals

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Konami GX687 / GX785

Sheet: /Timing/
File: timing.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/17

B_A2
74LS157
Quad Mux
AB[1..13]
AB12
AB13
GND
P2H
P2H
GND
A11
A12
B_B2
74LS157
Quad Mux
AB8
AB9
AB10
AB11
16V
32V
64V
128V
P2H
GND
A7
A8
A9
A10
B_B4B
74LS27
B_L6A
74LS04
FIXVRAMCS
P2H
MUWR
B_C2
74LS157
Quad Mux
AB4
AB5
AB6
AB7
64H
128H
256H
8V
P2H
GND
A3
A4
A5
A6
A[0..12]
B_B4A
74LS27
B_L6F
74LS04
FIXVRAMCS
P2H
MLWRD
B_D2
74LS157
Quad Mux
MREAD2D
AB1
AB2
AB3
8H
16H
32H
P2H
GND
OE
A0
A1
A2
10ns delay
50ns delay from OE active
to output data valid
7ns delay
B_J1A
74LS86
VCC
P4HD
P4H
B_A3
6264SL
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
VCC
GND
CS2
CS1
OE
WR
D8
D9
D10
D11
D12
D13
D14
D15
D[0..15]
B_B3
6264SL
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
VCC
GND
CS2
CS1
OE
WR
D0
D1
D2
D3
D4
D5
D6
D7
20ns delay
B_C4
74LS174
D9
D10
D11
D12
D13
D14
D0
D1
D2
D3
D4
D5
Cp
Mr
hex flip flop
VCC
Palette, HFLIP and VFLIP is valid for
eight pixels at a time and becomes valid
on the rising edge of P4H
B_C3
74LS174
D0
D1
D2
D3
D4
D5
Cp
Mr
hex flip flop
VCC
7ns delay
74LS86
B_J1C
VCC
P1HBD
P1HB
B_J2A
B_J1B
74LS86
P2H
P4H
HFLIPD
B_J2C
B_J2D
B_H2C
B_H2B
B_H2D
B_D3
74LS174
1V
2V
4V
D0
D1
D2
D3
D4
D5
Cp
Mr
hex flip flop
VCC
B_D4
74LS174
D3
D4
D5
D6
D7
D8
D0
D1
D2
D3
D4
D5
Cp
Mr
hex flip flop
VCC
RAM
A[0..5] Horizontal position 0..504 in steps of 8
A[6..10] Vertical position 0..248 in steps of 8
A[11.12] Tilemaps? Up to 4.
ROM
A[0..1] Horizontal position 0..6 in steps of 2
A[2..4] Vertical position 0..7
A[5..13] Tile number
B_D5
74LS245
DB15
DB14
DB13
DB12
DB11
DB10
DB9
DB8
A0
A1
A2
A3
A4
A5
A6
A7
B0
B1
B2
B3
B4
B5
B6
B7
buffers
A->B
CE
MREAD2D
D15
D14
D13
D12
D11
D10
D9
D8
{HVCOUNTERS}
DB[0..15]
B_E5
74LS245
DB7
DB6
DB5
DB4
DB3
DB2
DB1
DB0
D7
D6
D5
D4
D3
D2
D1
D0
B_C5A
74LS32
FIXVRAMCS
P2H
MREAD2D
D[0..8]: Tile number
D[9..12]: Palette
D13: HFLIP
D14: VFLIP
B_F3
27C128
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
GND
CE
OE
PGM
VPP
VCC
D0
D1
D2
D3
D4
D5
D6
D7
687M14/L14/N14
785H14/G14
Two pixels color data is output
on each rising edge of P1HB.
The current pixel data is multiplexed
with P1HB (and HFLIP).
B_E2
74LS273
D0
D1
D2
D3
D4
D5
D6
D7
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
flip flops
Cp
Mr
VCC
FX[20..23]
FX20
FX21
FX22
FX23
FX10
FX11
FX12
FX13
FX[10..13]
B_F2
74LS157
Quad Mux
FX10
FX11
FX12
FX13
FX20
FX21
FX22
FX23
GND
P1HBD
HFLIP
74LS86
B_J1D
B_H2A
The total delay for a pixel is seven pixel clocks
from when P2H and OE goes high and low respectively.
B_G3
74LS273
B_H3
74LS273
B_H4
74LS273
CLK21
VCC
FX0
FX1
FX2
FX3
FX4
FX5
FX6
FX7
FX[0..7]
FX[0..3]: Color
FX[4..7]: Palette
CLK21
P1H
P1HB
P1HB
P2H
P4H
P4H
1
2
3
4
5
6
7
8
3. Data out latched
into DFF's.
4. DFF's data
out valid
6. DFF's data
out valid
2. Data out from
RAM valid
4. FX[0..7] Data out valid
5. Palette and color values
latched into DFF's 3C and 2E.
1. Addresses and OE
latched into RAM. This condition sets the start:
P2H & P4H = 1.

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Fixed Layer/
File: fixed_layer.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/17

ZIP RAM

256kB tile RAM
16-bit access by CPU
32-bit access by video

This is from jammarcade.net

Konami 007644

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

CPUCTL flag must be set in 007779 to get the VA bus driven by the CPUs

VD[0..12]: Tile number
VD[13..15]: Palette Color

The 007781 is a shift register. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that. It reads 32 bits / 4 bits / pixel = 8 pixels at a time.

1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette

Data latched from the RAM:
Edge Tilemap
T1–T21
T2–T32

It is output 2 pixel clocks later to the ROMs.

When VA is switched to output the bus is updated for each P1H cycle.

The vertical counters are not the same value when they are offset with the scroll registers.

Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 –> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 –> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2

Outputs of RAM active when P4H is low, i.e during T1 and T2.

SYNCRLDV: Usually 60.604Hz
T is ~150ns.

VCNTEN: 8.0000kHz

HSYNC: 16.0000kHz, T2: 5.2175us

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Tile Layers/
File: tile_layers.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 10.0.4		Id: 7/17

B_K1
007783

B_L3
6264SL

B_M3
6264SL

B_K3
007784

74LS245

B_K5
007785

74LS157

B_C5B
74LS32

B_J8A
74LS32

B_H11A
74LS74

B_P1
PAL16L8'

B_RA2
2.2k x 8

B_RA3
2.2k x 8

Each unparsed sprite (object) occupies 40 words in object RAM. Only the first 8 words of each object data is used. The memory range for unparsed data is up 17FFh. That leaves room for 153 (153 x 40 −1 = 17E7h) objects per frame.

On PWB(A) 302190A CHAPAGE (CN2:A23) is connected to GND

007784 Sprite parser

− subtracts the scroll value for the sprites from current coordinates for the sprites.
− When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eigth words into four words and also subtract the x− and y−scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
− Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.

007793

CLK21′ = CLK21

$\overline{LRW}$ = -(DMAONS & OBJBUFINIT & CLK21′ |
  BEFLAG & DMA & P1H & DMAONS & CLK21′ & (-8H & HFLIP | 8H & -HFLIP)
 ~LWR & ~DMAINIT)

$\overline{URW}$ = -(DMAONS & OBJBUFINIT & CLK21′ |
  BEFLAG & DMA & P1H & DMAONS & CLK21′ & (8H & ~HFLIP | ~8H & HFLIP) |
  ~UWR & ~DMAINIT)

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG & DMA
3. ~DMAINIT

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Sprite Table/
File: sprite_table.kicad_sch

Title: Twin 16

Size: A3	Date: 2024−09−08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 8/17

Graphics Tile Data (ROM and RAM)

This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.

Konami 007788

The inverted ~P2H is written as N2H.

~OCRDL = ~(~SREAD & SAB1 & ~OCHARAMCS & OCHAROMCS | ~SREAD & SAB1 & ~OCHAROMCS)
OCWL = ~(~P2H & SREAD & SAB1 & ~OCHARAMCS)
OCRDH = ~(~SREAD & ~SAB1 & ~OCHARAMCS & OCHAROMCS | ~SREAD & ~SAB1 & ~OCHAROMCS)
OCWH = ~(~P2H & SREAD & ~SAB1 & ~OCHARAMCS)
VCWENL = ~(P2H & P4H & ~VCHARAMCS & SREAD & SAB1)
VCRDL = ~(~VCHARAMCS & ~SREAD & SAB1)
VCRDH = ~(~VCHARAMCS & ~SREAD & ~SAB1)
VCWENH = ~(P2H & P4H & ~VCHARAMCS & SREAD & ~SAB1)
P2HP4H = ~(P2H & P4H)
N2HP4H = ~(~P2H & P4H)

128kB RAM
for dynamic tiles

Pixel color data output rate is:
16-bits / 2 pixels = 8 bits / pixel.

W[15]: Vertical flip

Vertical flipped ROM's?

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Konami GX687 / GX785

Sheet: /Sprite ROM and RAM/
File: sprite_rom_and_ram.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 10.0.4		Id: 9/17

Frame buffer 1

2 x 256kB frame buffers
Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are reveresed for FB2.

Frame buffer 2

Not certain which of the parts 6KB,6UB and 6UD are used where. Some errors in the original schematics.

Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true
(R[0:3] != 4'h0) && (FBWM1 | FBWM2 | -WR | -VPD8)
data is written to DRAM.

Data is written when:
– WR is active.
– Not a transparent pixel (4'h0).
– FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.

OCD[0:3]: Pixel color
OCD[4:7]: Palette

Frame buffer 1 Palette color

Frame buffer 2 Palette color

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Konami GX687 / GX785

Sheet: /Sprite Frame Buffers/
File: sprite_frame_buffers.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 10.0.4		Id: 10/17

Components are on Video PCB

Components are on Main PCB

object layer:
0000: Transparent
1111: Shadow
else: Opaque

B_A11D 74LS32 · B_A11B 74LS32 · B_A11C 74LS32

FX2 12, FX3 13, FX1 9, FX0 10 → 11, 8 → 4, 5 → 6 FX

PI[0..2]

B_J8D 74LS32 · B_J8C 74LS32

1C3 12, 1C2 13, 1C1 9, 1C0 10 → 11, 8 → 4, 5 → 6 1C

B_J11 GAL22V10 007789

I1/CLK	1	23	SHADOW
PI0	2	22	
PI1	3	21	SELB
2C0	4	20	SELA
PI2	5	19	2C3
OCD0	6	18	2C2
OCD1	7	17	2C1
OCD2	8	16	
OCD3	9	15	
I10	10	14	
I11	11		
1C6	11		
2C6	13		
I13		24	VCC
		12	GND

{SEL}

Konami 007789

i11 (1C6) and i5 (PI2) is not used

~o15 = OCD0 & OCD1 & OCD2 & OCD3 | ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3

o16 = OCD0 & OCD1 & OCD2 & OCD3

SELA = FX +
/PI1 & PI0 & 1C +
/PI1 & OCD0 & OCD1 & OCD2 & OCD3 & 1C +
/PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & 1C +
PI1 & OCD0 & OCD1 & OCD2 & OCD3 & /2C0 & /2C1 & /2C2 & /2C3 +
PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & /2C0 & /2C1 & /2C2 & /2C3

SELB =/FX & OCD0 & OCD1 & OCD2 & OCD3 +
/FX & /OCD0 & /OCD1 & /OCD2 & /OCD3 +
/FX & PI0 & /PI1 & 1C +
/FX & PI0 & PI1 & 2C6 & 2C0 +
/FX & PI0 & PI1 & 2C6 & 2C1 +
/FX & PI0 & PI1 & 2C6 & 2C2 +
/FX & PI0 & PI1 & 2C6 & 2C3
o21.oe = vcc

/SHADOW = /FX & /PI1 & OCD0 & OCD1 & OCD2 & OCD3 +
/FX & /PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /1C +
/FX & PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C6 +
/FX & PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C0 & /2C1 & /2C2 & /2C3

B_H9 74LS153 Mux 4–1: I0a GND, I1a VCC, I2a GND, I3a; Ea GND; I0b VCC, I1b, I2b GND, I3b; Eb GND; SELA, SELB; S0, S1 → 7 COLOR8, 9 COLOR9

B_H8 74LS153 Mux 4–1: 6 2C6, 5 1C6, 4 OCD6, 3 FX6; Ea GND; I0b VCC, I1b GND, I2b 12, I3b 13; Eb GND; SELA, SELB; S0, S1 → 7 COLOR6, 9 COLOR7

B_H10 74LS153 Mux 4–1: 6 2C4, 5 1C4, 4 OCD4, 3 FX4; Ea GND; I0b, I1b 11, I2b 12, I3b 13; Eb GND; SELA, SELB; S0, S1 → 7 COLOR4, 9 COLOR5

B_J9 74LS153 Mux 4–1: 6 2C2, 5 1C2, 4 OCD2, 3 FX2; Ea GND; I0b, I1b 11, I2b 12, I3b 13; Eb GND; SELA, SELB; S0, S1 → 7 COLOR2, 9 COLOR3

B_J10 74LS153 Mux 4–1: 6 2C0, 5 1C0, 4 OCD0, 3 FX0; Ea GND; I0b, I1b 11, I2b 12, I3b 13; Eb GND; SELA, SELB; S0, S1 → 7 COLOR0, 9 COLOR1

2C[0..6], 1C[0..6], OCD[0..7], FX[0..7]

COLOR[0..9]

AB[1..12]

A_U6 74LS273 flip flops: COLOR2–9 → D0–D7; CLK2D 11, VCC 1, Cp, Mr; Q0–Q7 → 2 CB2, 5 CB3, 6 CB4, 9 CB5, 12 CB6, 15, 16, 19

A_T6 74LS157 Quad Mux: AB9 2, AB10 5, AB11 11, AB12 14 → I0a–I0d; I1a 3, I1b 6, I1c 10, I1d 13; GND; COLORRAM 1, GND 15 (SELECT_A–B, STROBE_G); Za 4 C9, Zb 7 C8, Zc 9 C7, Zd 12 C6

A_U7 74LS174 hex flip flop: 3 CBLK, 6, 11, 14 COLOR0–1; CLK2D 9, VCC 1, Mr; D0–D5; Q0–Q5 → 2, 5, 7, 10, 12 CB0, 15 CB1

CBLK'

CB[0..6]

AB1 3 → A_F7B 4 (74LS04)

A_U1 007327: C6 20, C7 19, C8 18, C9 17, AB8 29, AB7 30, AB6 31, AB5 32, AB4 33, AB3 34, AB2 35, AB1 37, A0 36 → RA11, RA10, RA9, RA8, A7, A6, A5, A4, A3, A2, A1, A0, A0; CB6 22, CB5 23, CB4 24, CB3 25, CB2 26, CB1 27, CB0 28 → CB6, CB5, CB4, CB3, CB2, CB1, CB0

DB[0..7]: DB7 3, DB6 4, DB5 5, DB4 6, DB3 7, DB2 8, DB1 9, DB0 10 → D7, D6, D5, D4, D3, D2, D1, D0

CLK2D 40 → CLK; MREAD1 38 → R/W; COLORRAM 21 → SEL; 1 → CCS; MLWRR 2 → CWR; 39 → BLK

VCC 12 → VCC; GND 11, 16 → GND

R 15 → RED; G 14 → GREEN; B 13 → BLUE

A_R3 82, A_R4 82, A_R5 82

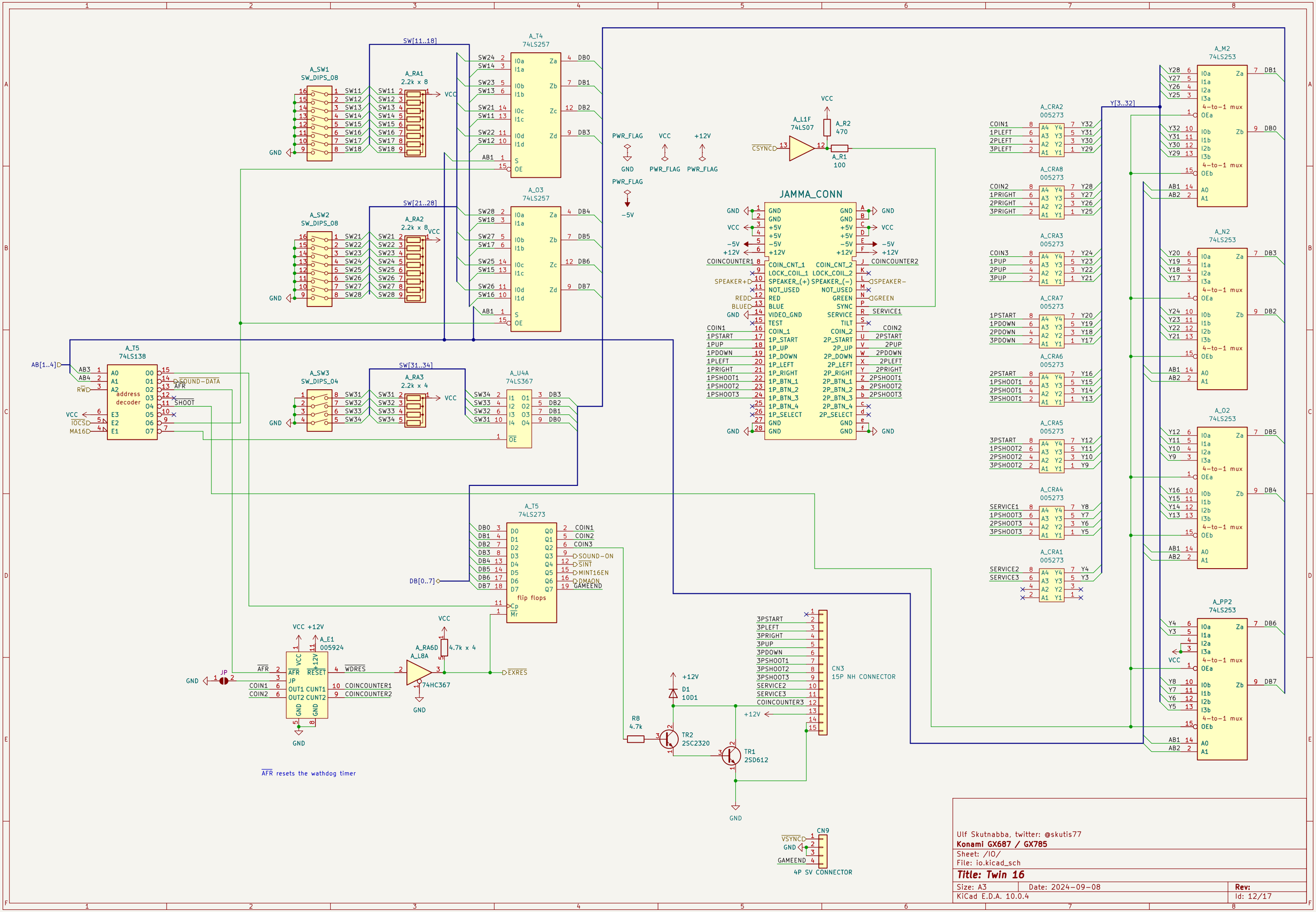
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Color mixer/
File: color_mixer.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 11/17



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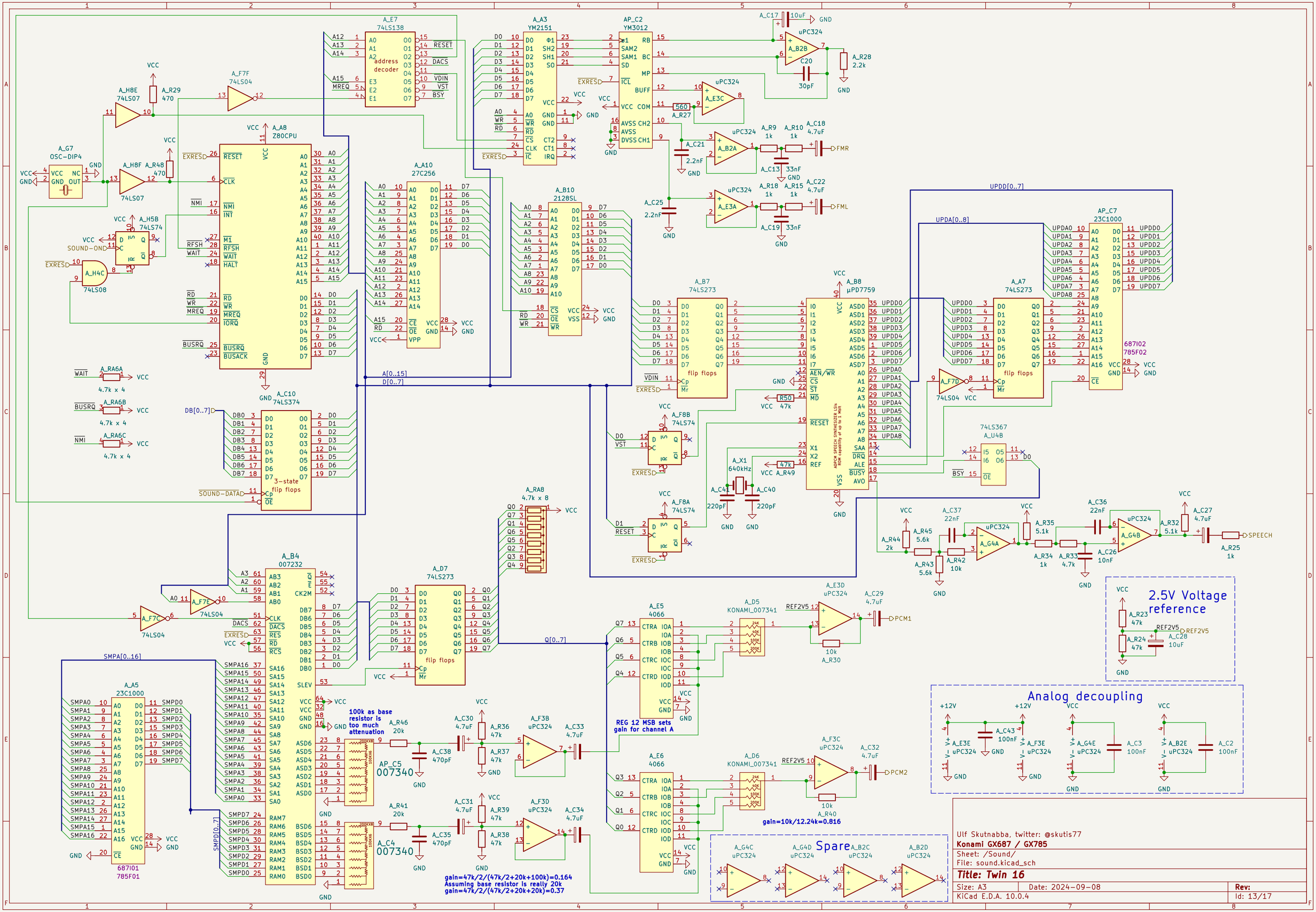
Konami GX687 / GX785

Sheet: /IO/

File: io.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 12/17


A_E7
74LS138
address decoder
A_A3
YM2151
AP_C2
YM3012
A_C17 10uF
uPC324
A_R28 2.2k
C20 30pF
A_H8E
74LS07
A_R29 470
A_F7F
74LS04
A_G7
OSC-DIP4
A_H8F
A_R48 470
A_A8
Z80CPU
A_A10
27C256
A_B10
2128SL
A_H5B
74LS74
A_H4C
74LS08
A_R9 1k
A_R10 1k
A_C18 4.7uF
A_C21 2.2nF
A_C13 33nF
A_C22 4.7uF
A_C25 2.2nF
A_C19 33nF
A_R18 1k
A_R15 1k
A_B7
74LS273
A_B8
µPD7759
A_A7
74LS273
AP_C7
23C1000
UPDD[0..7]
UPDA[0..8]
A_F7D
74LS04
687I02
785F02
A_RA6A
4.7k x 4
A_RA6B
4.7k x 4
A_RA6C
4.7k x 4
A_C10
74LS374
3-state flip flops
A_F8B
74LS74
A_F8A
74LS74
A_X1
640kHz
A_C41 220pF
A_C40 220pF
A_R49 47k
R50 47k
74LS367
A_U4B
A_RA8
4.7k x 8
A[0..15]
D[0..7]
A_B4
007232
A_D7
74LS273
Q[0..7]
SMPA[0..16]
A_A5
23C1000
687I01
785F01
SMPD[0..7]
100k as base resistor is too much attenuation
AP_C5
007340
A_C4
007340
A_R46 20k
A_R41 20k
A_C38 470pF
A_C35 470pF
A_C30 4.7uF
A_C31 4.7uF
A_R36 47k
A_R37 47k
A_R39 47k
A_R38 47k
A_F3B
uPC324
A_F3D
uPC324
A_C33 4.7uF
A_C34 4.7uF
gain=47k/2/(47k/2+20k+100k)=0.164
Assuming base resistor is really 20k
gain=47k/2/(47k/2+20k+20k)=0.37
REG 12 MSB sets gain for channel A
A_E5
4066
A_E6
4066
A_D5
KONAMI_007341
A_D6
KONAMI_007341
A_E3D
uPC324
A_F3C
uPC324
A_C29 4.7uF
A_C32 4.7uF
A_R30 10k
A_R40 10k
PCM1
PCM2
gain=10k/12.24k=0.816
A_R44 2k
A_R45 5.6k
A_R43 5.6k
A_C37 22nF
A_R42 10k
A_G4A
uPC324
A_R35 5.1k
A_R34 1k
A_R33 4.7k
A_C36 22nF
A_C26 10nF
A_G4B
uPC324
A_R32 5.1k
A_C27 4.7uF
A_R25 1k
SPEECH
2.5V Voltage reference
A_R23 47k
A_R24 47k
A_C28 10uF
REF2V5
Analog decoupling
A_C43 100nF
A_C3 100nF
A_C2 100nF
Spare
A_G4C
A_G4D
A_B2C
A_B2D
uPC324
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Sound/
File: sound.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
KiCad E.D.A. 10.0.4
Rev:
Id: 13/17

To enable stereo sound output on connector CN8:
Use the connector which shortens Pin 1 -> Pin 2 and
Pin 3 -> Pin 4 on Connector CN7.

A_R17
1k
FMRD
A_R12
7.5k
SPEECHD
A_R11
5.1k
PCM1D
A_R19
5.1k
PCM2D
A_R16
1k
FMLD
A_R22
7.5k
SPEECHD
A_R31
5.1k
PCM1D
A_R20
5.1k
PCM2D
A_E3B
uPC324
REF2V5D
A_R13
12k
+12V
A_R52
3.3k
A_C15
4.7uF
A_R14
10k
A_F3A
uPC324
REF2V5D
A_R21
12k
+12V
A_R53
3.3k
A_C16
4.7uF
A_R26
10k
CN6
4P NH CONN
GND
CN7
4P NH CONN
APOT_R53
1k
APOT_R52
1k
A_C11
100nF
A_C12
100nF
GND
A_C9
100uF
ACAP_C10
100uF
A_D1
MB3722
IN-A
IN+A
IN+A
IN-B
VCC
GNDIN
GNDOUT
BYPASS
BSA
OUTA
BSB
OUTB
+12V
ACAP_C3
1000uF
A_C14
10uF
A_C8
100uF
A_C7
100uF
ACAP_C2
1000uF
A_C6
100nF
A_R6
1
A_C5
100nF
A_R7
1
A_C1
1000uF
SPEAKER-
SPEAKER+
CN8
4P NH CONN

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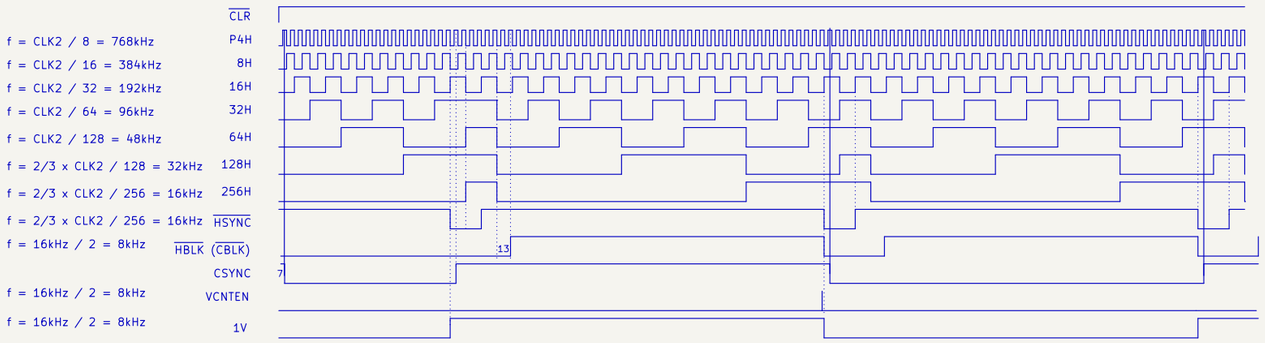
Konami GX687 / GX785

Sheet: /Sound mixer/
File: sound_mixer.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 14/17

Horizontal signals

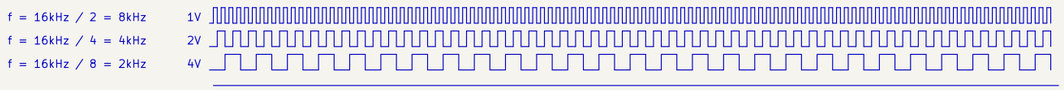


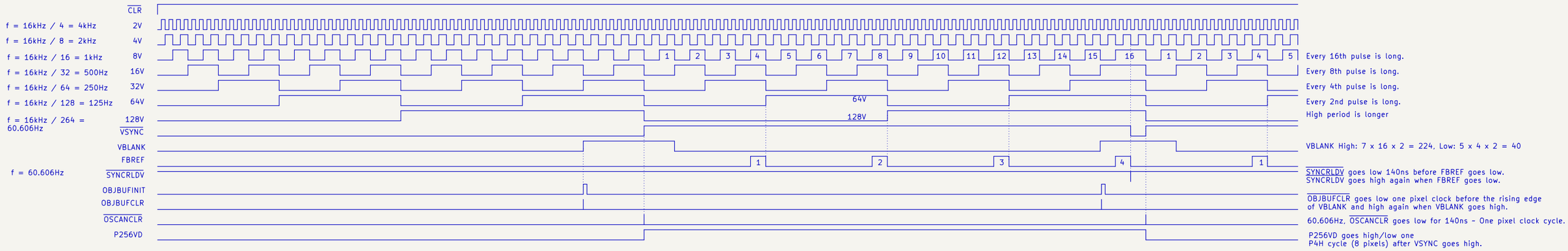
The first HSYNC is at HCOUNT 175 after reset.

– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

Vertical signals





FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

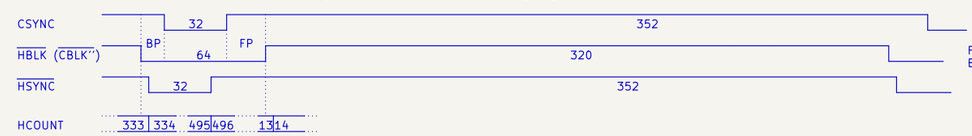
Horizontal and vertical synch timing diagrams

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

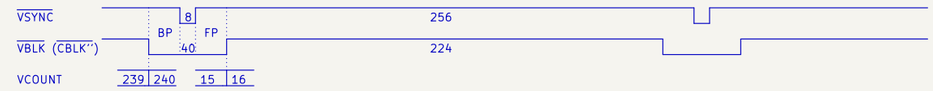
Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

CBLK'' is at the output of color mixer 007327.
CBLK' is shifted inside the 007327 one pixel clock.

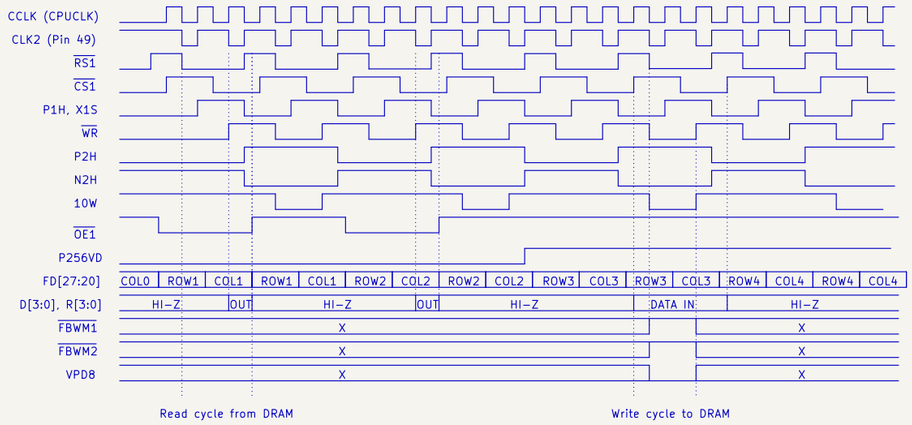
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Konami GX687 / GX785

Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 15/17



The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

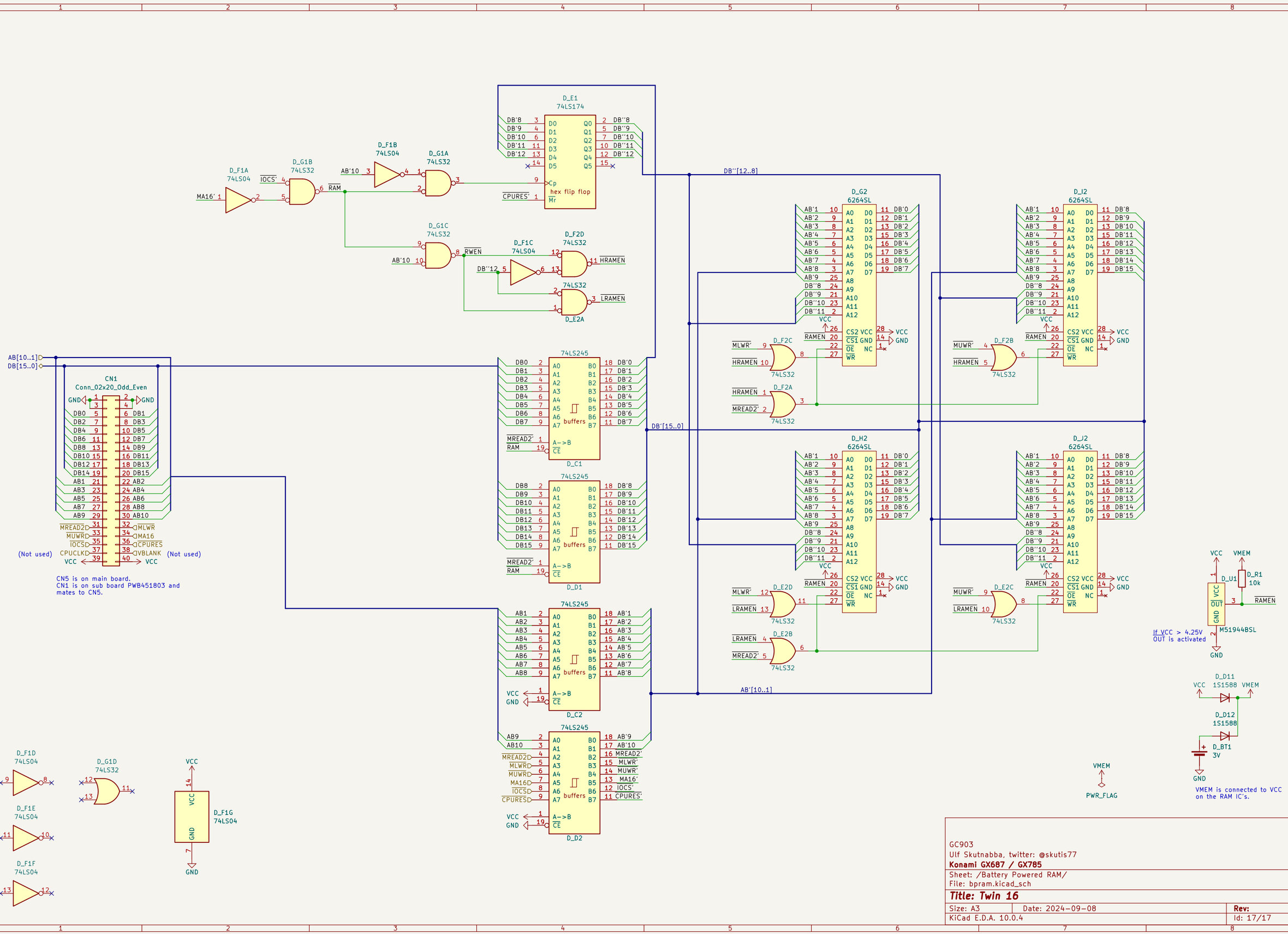
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Sprite timing diagrams/
File: sprite_timing_diagrams.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 16/17


D_E1
74LS174
DB'8 3
DB'9 4
DB'10 6
DB'11 11
DB'12 13
D0
D1
D2
D3
D4
D5
Q0
Q1
Q2
Q3
Q4
Q5
2 DB''8
5 DB''9
7 DB''10
10 DB''11
12 DB''12
14
15
9
Cp
hex flip flop
Mr
CPURES' 1
D_F1A
74LS04
IOCS' 4
MA16' 1
2
5
D_G1B
74LS32
6 RAM
AB'10 3
D_F1B
74LS04
4
D_G1A
74LS32
1
2
3
D_G1C
74LS32
9
8 RWEN
AB'10 10
D_F1C
74LS04
12
13
6
D_F2D
74LS32
11 HRAMEN
74LS32
2
1
3 LRAMEN
D_E2A
DB''12 5
DB''[12..8]
D_G2
6264SL
AB'1 10
AB'2 9
AB'3 8
AB'4 7
AB'5 6
AB'6 5
AB'7 4
AB'8 3
AB'9 25
DB''8 24
DB''9 21
DB''10 23
DB''11 2
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
D0
D1
D2
D3
D4
D5
D6
D7
11 DB'0
12 DB'1
13 DB'2
15 DB'3
16 DB'4
17 DB'5
18 DB'6
19 DB'7
VCC
26
CS2 VCC
CS1 GND
OE NC
WR
RAMEN 20
22
27
28 VCC
14 GND
1
D_F2C
MLWR' 9
HRAMEN 10
8
74LS32
D_F2A
HRAMEN 1
MREAD2' 2
3
74LS32
D_J2
6264SL
D_F2B
MUWR' 4
HRAMEN 5
6
74LS32
D_H2
6264SL
D_E2D
MLWR' 12
LRAMEN 13
11
74LS32
D_E2C
MUWR' 9
LRAMEN 10
8
74LS32
D_E2B
LRAMEN 4
MREAD2' 5
6
74LS32
AB[10..1]
DB[15..0]
CN1
Conn_02x20_Odd_Even
GND 1
3
2
4 GND
DB0 5
DB2 7
DB4 9
DB6 11
DB8 13
DB10 15
DB12 17
DB14 19
6 DB1
8 DB3
10 DB5
12 DB7
14 DB9
16 DB11
18 DB13
20 DB15
AB1 21
AB3 23
AB5 25
AB7 27
AB9 29
AB10 31
22 AB2
24 AB4
26 AB6
28 AB8
30
MREAD2 33
MUWR 35
IOCS 37
CPUCLK 39
32 MLWR
34 MA16
36 CPURES
38 VBLANK
40 VCC
(Not used)
(Not used)
VCC
CN5 is on main board.
CN1 is on sub board PWB451803 and
mates to CN5.
74LS245
DB0 2
DB1 3
DB2 4
DB3 5
DB4 6
DB5 7
DB6 8
DB7 9
A0
A1
A2
A3
A4
A5
A6
A7
buffers
B0
B1
B2
B3
B4
B5
B6
B7
18 DB'0
17 DB'1
16 DB'2
15 DB'3
14 DB'4
13 DB'5
12 DB'6
11 DB'7
MREAD2' 1
RAM 19
A->B
CE
D_C1
74LS245
DB8 2
DB9 3
DB10 4
DB11 5
DB12 6
DB13 7
DB14 8
DB15 9
18 DB'8
17 DB'9
16 DB'10
15 DB'11
14 DB'12
13 DB'13
12 DB'14
11 DB'15
D_D1
74LS245
AB1 2
AB2 3
AB3 4
AB4 5
AB5 6
AB6 7
AB7 8
AB8 9
18 AB'1
17 AB'2
16 AB'3
15 AB'4
14 AB'5
13 AB'6
12 AB'7
11 AB'8
VCC 1
GND
19
D_C2
74LS245
AB9 2
AB10 3
MREAD2D 4
MLWRD 5
MWRD 6
MA16D 7
IOCSD 8
CPURESD 9
18 AB'9
17 AB'10
16 MREAD2'
15 MLWR'
14 MUWR'
13 MA16'
12 IOCS'
11 CPURES'
D_D2
D_F1D
74LS04
9
8
D_G1D
74LS32
12
13
11
D_F1E
74LS04
11
10
D_F1F
74LS04
13
12
VCC
14
VCC
GND
7
GND
D_F1G
74LS04
VCC
VMEM
D_U1
1
VCC
OUT
GND
2
3
D_R1
10k
RAMEN
M51944BSL
If VCC > 4.25V
OUT is activated
GND
D_D11
VCC
1S1588
VMEM
D_D12
1S1588
D_BT1
3V
GND
VMEM
PWR_FLAG
VMEM is connected to VCC
on the RAM IC's.
GC903
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Battery Powered RAM/
File: bpram.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 10.0.4
Id: 17/17